THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 2

Chapter 9: **The Amazing World of Solutes, Solvents and Solutions** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A mixture that is cloudy and settles on standing is: 1

(a) an element **(b)** a colloid **(c)** a suspension **(d)** a true solution

2. A saturated solution of sugar at 30 degrees Celsius is heated to 60 degrees Celsius. It becomes: 1

(a) unsaturated **(b)** a colloid **(c)** a suspension **(d)** saturated

3. What mass of solute is in 60 g of 25 per cent solution? 1

(a) 35 g **(b)** 15 g **(c)** 25 g **(d)** 60 g

4. When potassium permanganate is added to water the solution is: 1

(a) colourless **(b)** purple and homogeneous
(c) black **(d)** heterogeneous

5. Which one will dissolve the most CO2 per litre of water? 1

(a) hot water in open air **(b)** cold water under pressure
(c) warm water in open air **(d)** boiling water

6. A 20 per cent solution contains 10 g of solute. The mass of the solution is: 1

(a) 50 g **(b)** 200 g **(c)** 10 g **(d)** 20 g

7. When the saturated solution of salt at 50 degrees Celsius is cooled, some salt: 1

(a) evaporates **(b)** boils
(c) comes out as crystals **(d)** dissolves more

8. A student dissolves 5 g of salt in 100 g of water. The amount of water that must be evaporated to get back the salt is: 1

(a) 25 g **(b)** half **(c)** all of it **(d)** 5 g

9. A solution is diluted by adding water. The mass percentage of the solute: 1

(a) decreases **(b)** is unchanged **(c)** becomes zero **(d)** increases

10. A cold saturated solution of sugar is warmed. More sugar can then be dissolved because: 1

(a) water increases **(b)** evaporation
(c) stirring **(d)** solubility increases with temperature

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Why does the pressure cap of a soft drink bottle hiss when opened? 2

12. How can you prove that a solution of sugar still contains sugar after it appears 'disappeared'? 2

13. Why are the particles of a true solution not visible even with a microscope? 2

14. Give two differences between dissolving and melting. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Explain how to prepare 200 g of 15 per cent salt solution. 3

16. Why does hot tea become sweeter faster with powdered sugar than with sugar lumps? 3

17. State three differences between dissolving sugar in water and mixing sand in water. 3

OR

A student mixes 50 g of a 20 per cent solution with 50 g of water. Find the new mass percentage and say what has happened to concentration. 3

18. A student makes solutions A, B, C by dissolving 5 g, 10 g, 15 g of salt in 100 g water. Which is most concentrated? Give the percentage for A. 3

OR

How does the shape of the solubility curve help a chemist? Explain with a statement about KNO3. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Explain the difference between true solutions, colloids and suspensions in particle size, filtration, settling, Tyndall effect and give two examples each. 5

OR

Explain why the volume of the solution is not equal to the sum of the volumes of solute and solvent in some cases but the mass is. Give an example. 5

20. Explain why water is essential for life as a solvent for gases, solids and liquids, and discuss the effect of pollution on dissolved oxygen. 5

OR

A 1 litre bottle of soda is opened. Explain the sequence of events: hissing, bubbles and flat taste. 5

END OF QUESTION PAPER — Total: 40 Marks

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